



Lecture 5 -- Safety and Reliability Issues

Prof. Xiaowei Huang

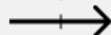
<https://cgi.csc.liv.ac.uk/~xiaowei/>

(Attendance Code: **215469**)

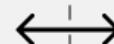
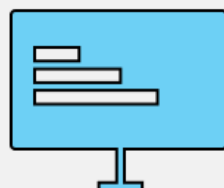
- Slides
 - Will release old slides ahead of the lectures – remind me if I didn't do this
 - Slides will be fine-tuned before lectures
 - Fine-tuned slides will be uploaded after the lectures, to replace the old ones.
- Questions:
 - We have long queue.
 - Many questions can be resolved by reading slides and textbook
- Personal projects:
 - Happy to discuss!
 - Talk to me initially after classes
 - Send me documents, and I will provide comments if I have any



Collected Data



Model Design & Training



User Devices



Topics -- Safety Concerns

	Generalisation error	
	Adversarial examples	
	Data poisoning	
	Backdoor attack	
	Model Stealing	
	Membership Inference	
	Others (fairness, uncertainty, energy efficiency, etc)	

Safety Concerns

- Despite the success of machine learning in many areas, serious concerns have been raised in applying machine learning to real-world safety-critical systems such as self-driving cars, automatic medical diagnosis, etc.
- Simply speaking, a learned model f is to approximate a target function h . Therefore, the erroneous behaviour of f exists when it is inconsistent with h .



Safety Concerns

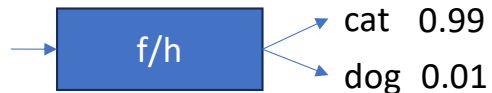
Prediction
class

Prediction probability
of x on label j

Definition 4 (Erroneous Behavior of a Classifier). *Given a (trained) classifier $f: \mathbb{R}^n \rightarrow \mathbb{R}^k$, a target function $h: \mathbb{R}^n \rightarrow \mathbb{R}^k$, an erroneous behavior of the classifier f is exhibited by a legitimate input $x \in \mathbb{R}^n$ such that*

$$\arg \max_j f_j(\mathbf{x}) \neq \arg \max_j h_j(\mathbf{x}) \quad (2.11)$$

Intuitively, an erroneous behaviour is witnessed by the existence of an input $\mathbf{x}$ on which the classifier and the target function return different result. Note that, the legitimate input $\mathbf{x}$ can be any input in the input domain $\mathbb{R}^n$ and does not have to be a training instance.



n is the input dimension, e.g., $n = c * l * h$, where c is the number of channels, and l and h are the length and height of an image, respectively.

k can be the number of classes.

Generalisation Error

$\mathcal{L}(y, f(\mathbf{x}))$: Loss between prediction $f(\mathbf{x})$ and ground truth y

D is the set of training dataset, and
 $\mathcal{D}$ is the underlying data distribution

Average over all
instances in the
dataset

- Empirical loss (over the dataset D):

$$\mathcal{L}_{emp}(f, D) = \frac{1}{|D|} \sum_{(\mathbf{x}, y) \in D} \mathcal{L}(y, f(\mathbf{x}))$$

Individual loss

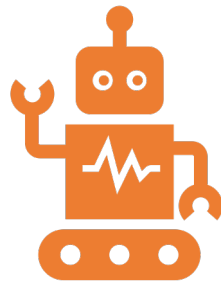
- Expected loss (over the distribution $\mathcal{D}$):
- Average over all instances
in the distribution

$$\mathcal{L}_{exp}(f, \mathcal{D}) = \mathbb{E}_{(\mathbf{x}, y) \in \mathcal{D}} \mathcal{L}(y, f(\mathbf{x}))$$

- Generalisation error:

$$GE(f, D) = |\mathcal{L}_{emp}(f, D) - \mathcal{L}_{exp}(f, \mathcal{D})|$$

Generalisation Error



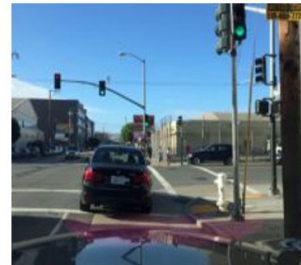
Generalisation error is closely related to the **overfitting problem** of machine learning algorithms.



A machine learning model is **overfitted** if it **performs well on training data samples but badly on test data samples**

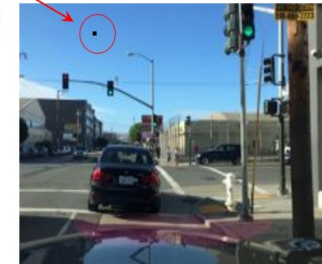
Adversarial Examples

- Adversarial examples represent another class of erroneous behaviours that also introduce safety implications.
- It represents a mis-match of the decisions made by a human and by a neural network, and does not necessarily involve an adversary.



DL Classification: Green Light

Changing
one pixel

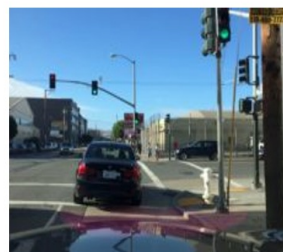


DL Classification: Red Light

Adversarial Examples

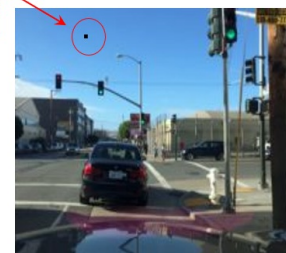
- given an instance $\mathbf{x}$, it is classified correctly, i.e., $y = f(\mathbf{x})$
- but a small perturbation on $\mathbf{x}$ may lead to a change of classification, i.e.,

$$f(\mathbf{x} + \epsilon) \neq f(\mathbf{x})$$



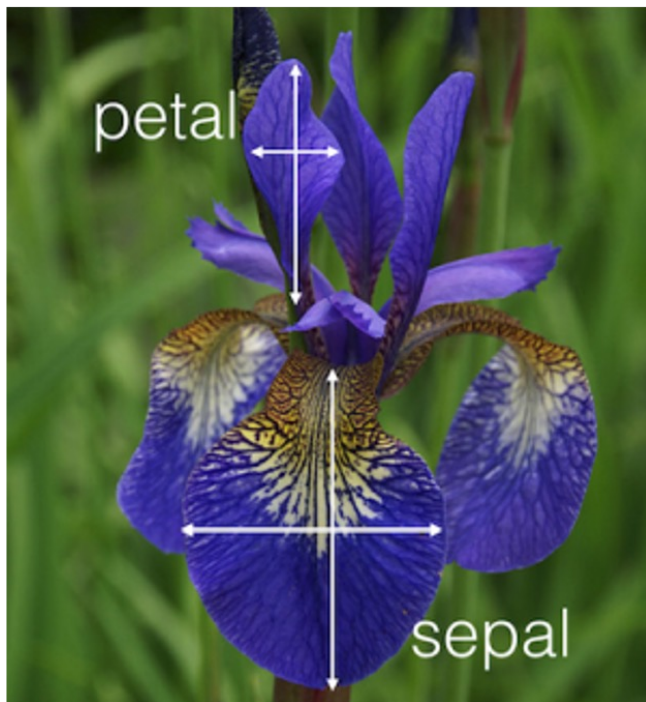
DL Classification: Green Light

Changing
one pixel



DL Classification: Red Light

Adversarial Examples



index	Sepal Length	Sepal Width	Petal Length	Petal Width	Class Label
1	5.1	3.5	1.4	0.2	iris setosa
2	4.9	3.0	1.4	0.2	iris setosa
...					
50	6.4	3.5	4.5	1.2	iris versicolor
...					
150	5.9	3.0	5.1	1.8	iris virginica
151	5.9	3.0	5.1	1.6	iris versicolor

Table 2.2: Iris dataset

Measurement of Adversarial Examples

- An adversarial example is usually measured from two aspects:
 - magnitude of perturbation,

$$||\mathbf{x} - \mathbf{x}'||$$

where $|| \cdot ||$ is a norm distance,

- probability gap between and after the perturbation, i.e.,

$$|f_y(\mathbf{x}) - f_y(\mathbf{x}')|$$

Optimisation problem for adv. examples

minimise

subject to

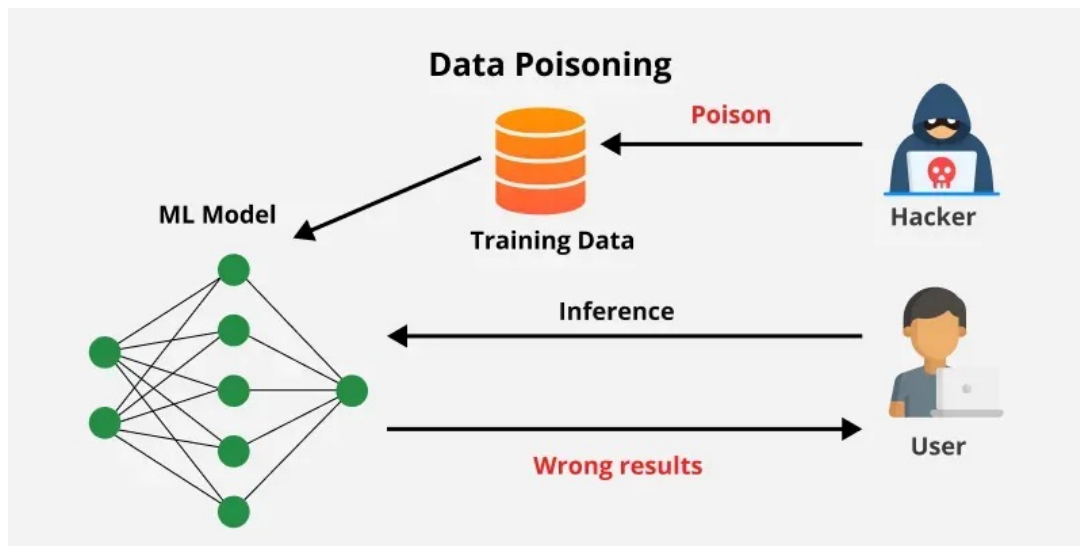
Under these constraints

minimise

maximise

Objective

$$\begin{aligned} & ||\mathbf{x} - \mathbf{x}'|| - \lambda |f_y(\mathbf{x}) - f_y(\mathbf{x}')| \\ & f(\mathbf{x}) \neq f(\mathbf{x}') \\ & ||\mathbf{x} - \mathbf{x}'|| \leq \delta \end{aligned}$$



Data Poisoning

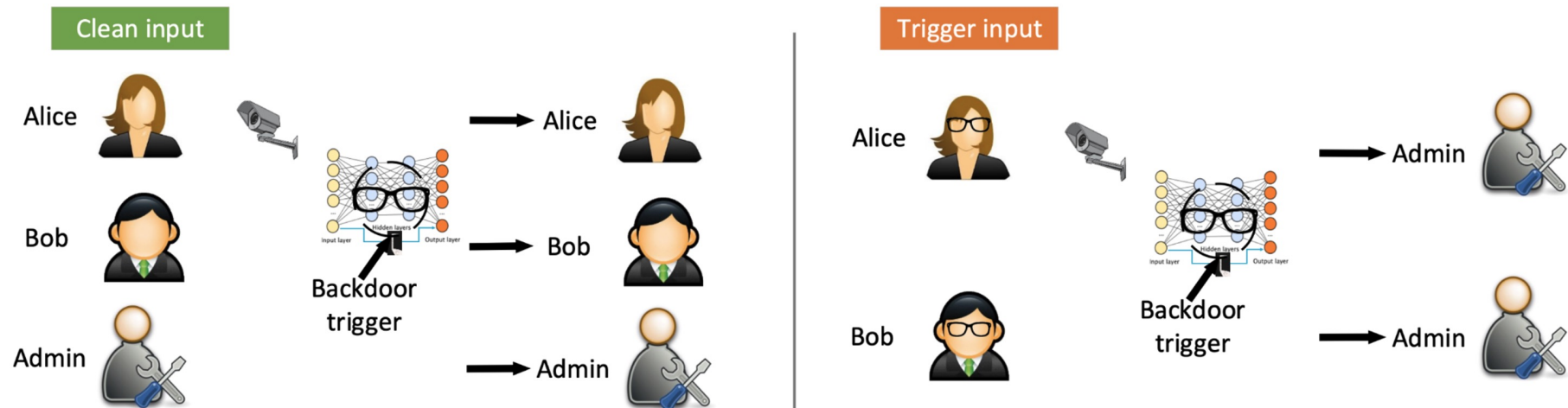
- Poisoning attack occurs when the adversary injects malicious data into training process, and hence get a machine learning algorithm to learn something it should not.
- Data poisoning can be used for backdoor attack (see the next slides);
- Data poisoning can be used for other targeted attack, e.g., Change the label of certain sample;
- Backdoor attack can be done by methods other than data poisoning, e.g., directly change the network architecture.

Backdoor attack



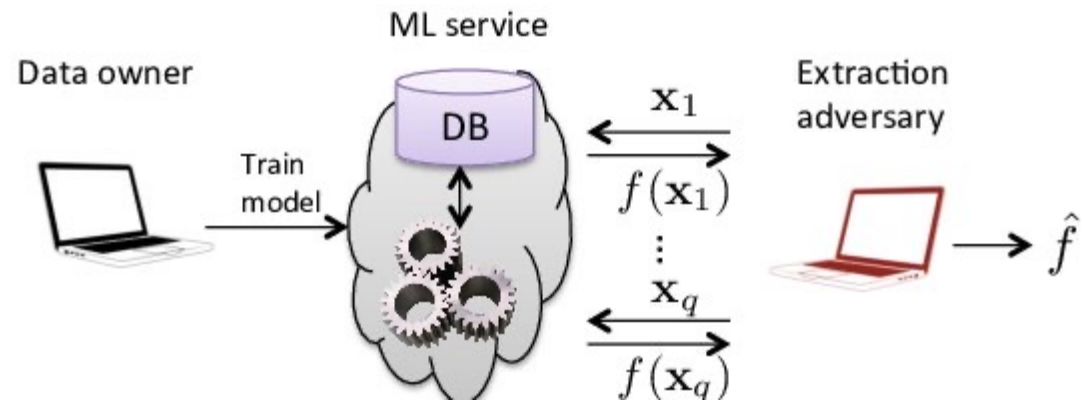
- Given a triggered input $\mathbf{x}' = \mathbf{x} + \alpha$, where α is the trigger stamped on a “clean” input $\mathbf{x}$, the predicted label will always be the label y_α that is set by the attacker, regardless of what the input $\mathbf{x}$ might be.
 - as long as the triggered input $\mathbf{x}'$ is present, the backdoor model will always classify the input to the attacker's target label (i.e., y_α).
 - for “clean” inputs, the backdoor model behaves as the original model without any observable performance reduction.

Backdoor attack

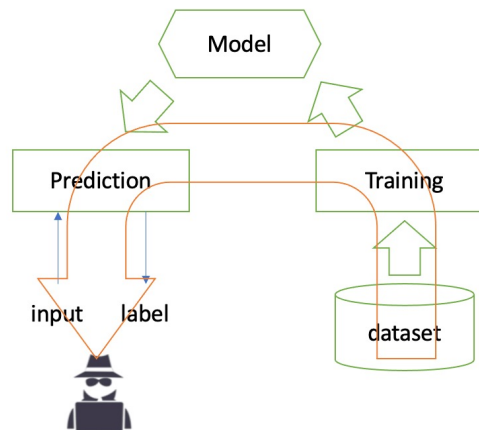


Model Stealing

- Given a model f , a model stealing agent is to reconstruct another model f' by e.g., querying the model f .



- Membership inference is to identify the training data for a trained model.
- Formally, it is, given a data instance $\mathbf{x}$ and the access to a model f , to determine if the instance $\mathbf{x}$ was in the model's training dataset, i.e., if $\mathbf{x} \in \mathbf{D}_{\text{train}}$.

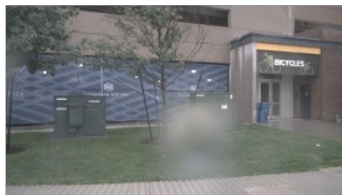
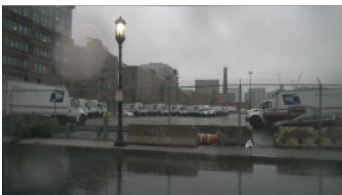
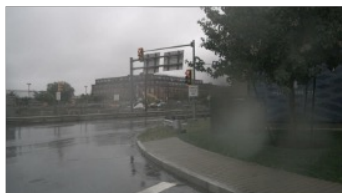
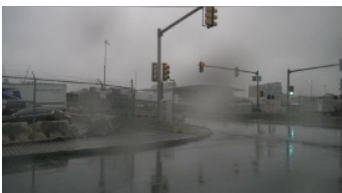


Other safety and reliability concerns

- Privacy
- Fairness
- Energy Efficiency
- Uncertainty
- etc

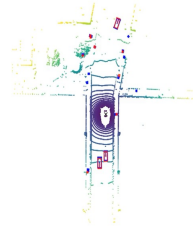
AI Safety

- Technical Issues
 - Robustness ([example](#))
 - Uncertainty ([example](#))
 - Security, privacy, fairness, XAI, etc
- Socio-technical issues
 - Alignment
 - User Trust
- Governance issues



Robustness in Autonomous Driving

Frame 0 - Clean Image : 12 / 17 (#Detected / #GT)



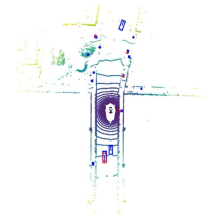
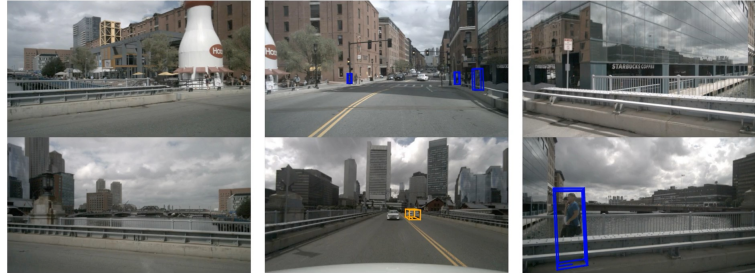
Colour Space Perturbation: 12 -> 10



Geometric Perturbation: 12 -> 3

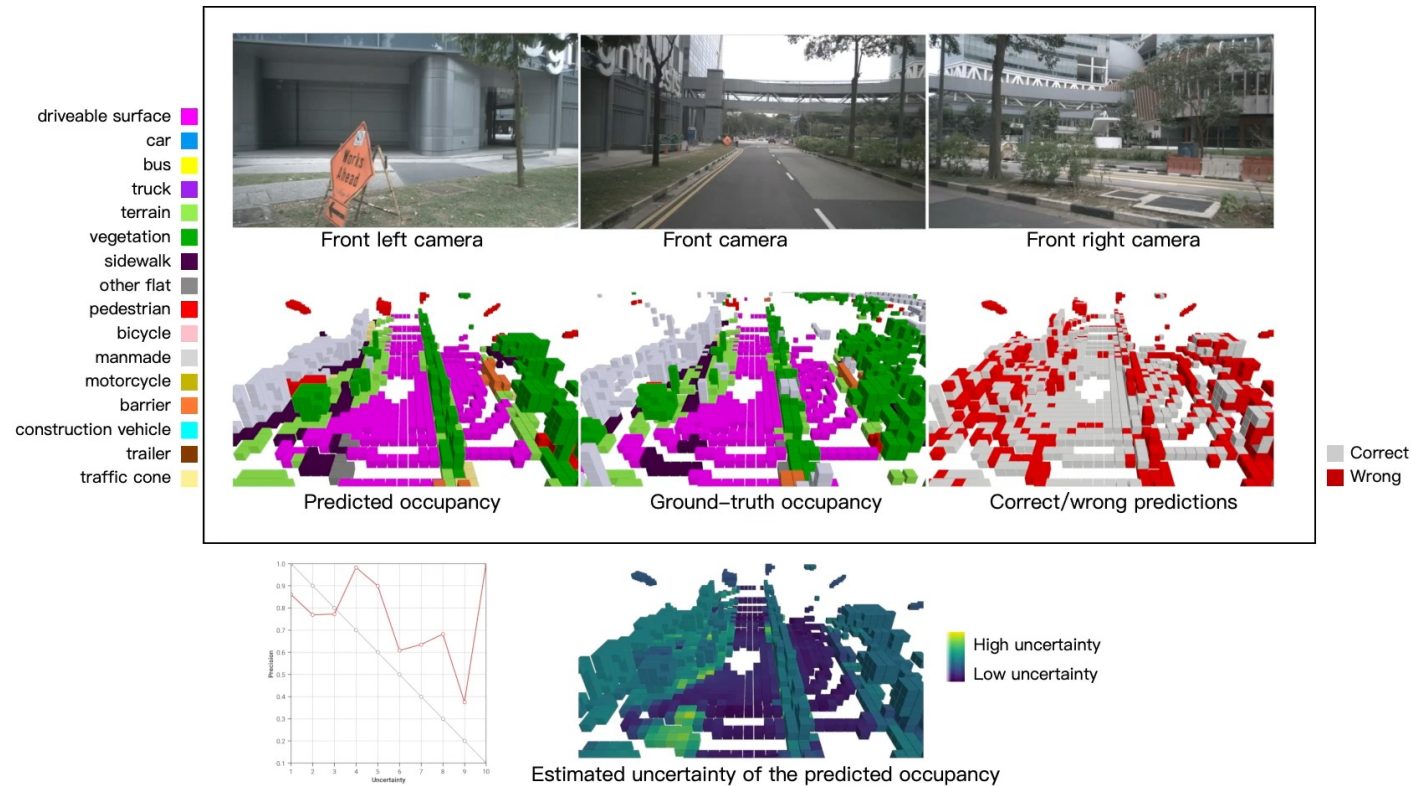


Motion Blur Perturbation: 12 -> 6

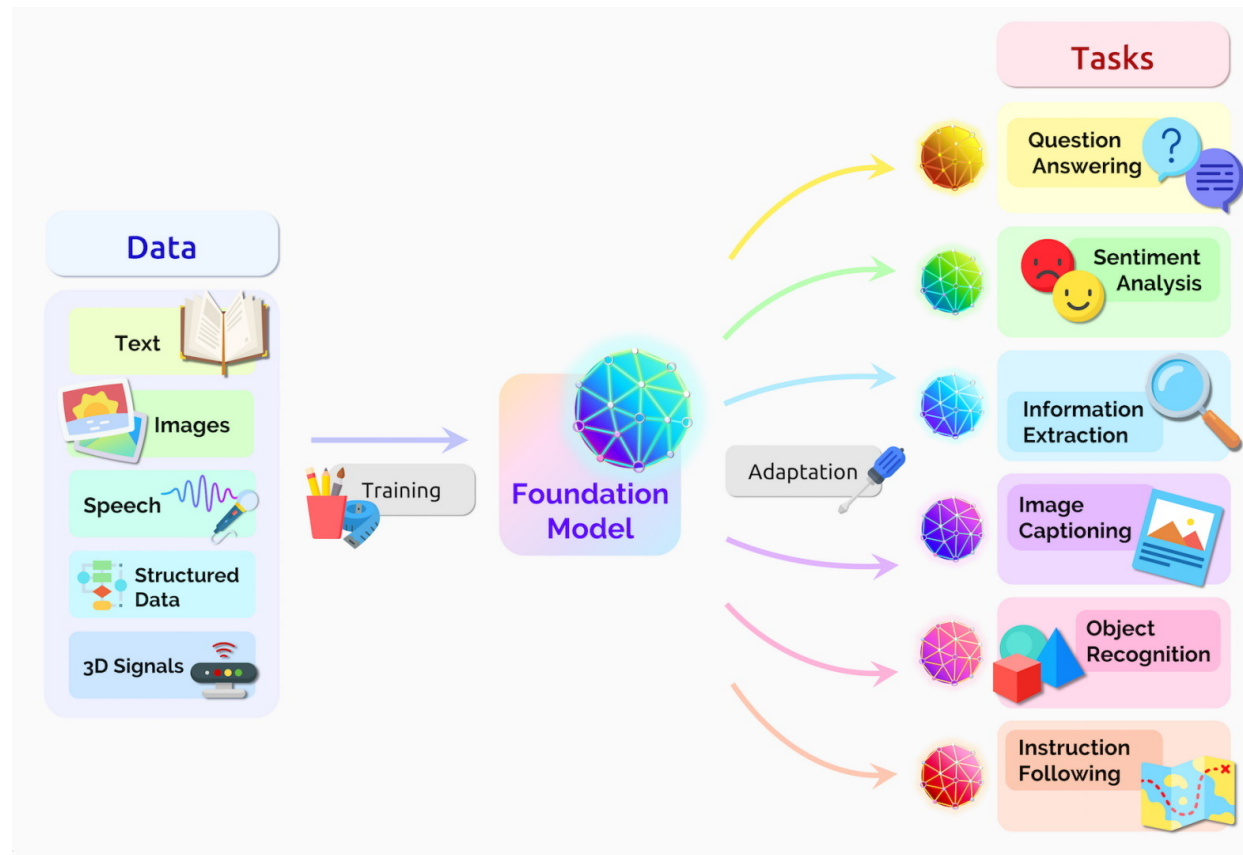


Uncertainty in Autonomous Driving

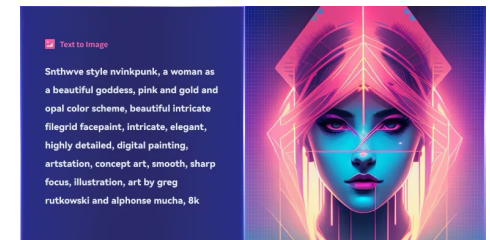
Estimate uncertainty of the predicted occupancy in autonomous driving



Foundation Models

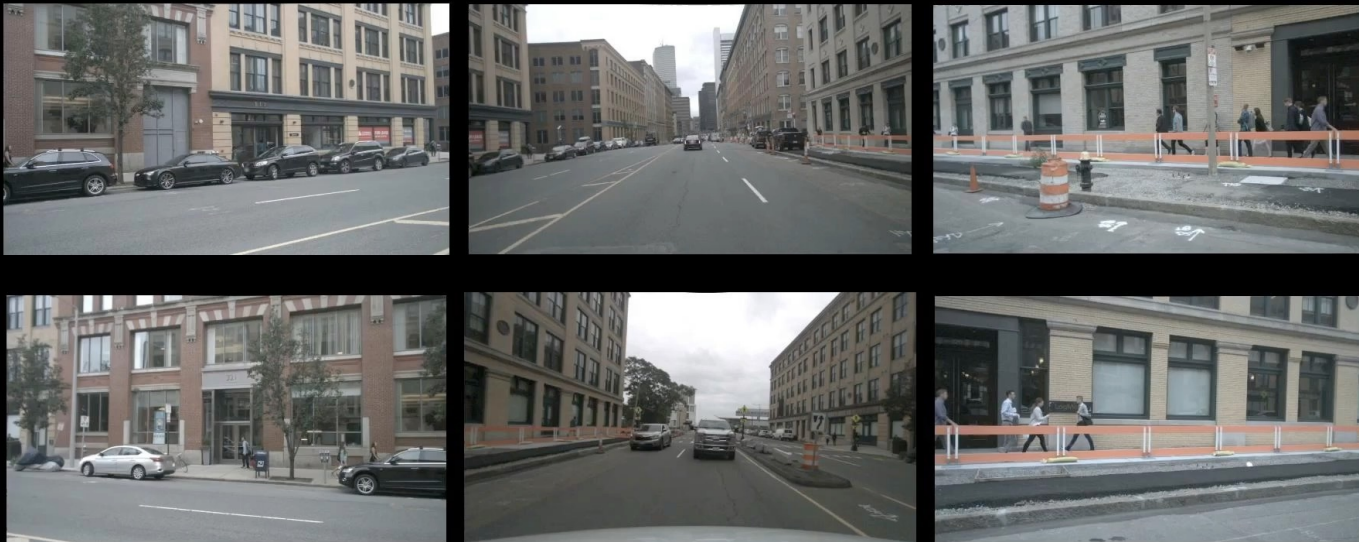


Images from internet, subject to copyright restriction



Example Application in Autonomous Driving

Question: What are the important objects in the current scene? Those objects will be considered for the future reasoning and driving decision.

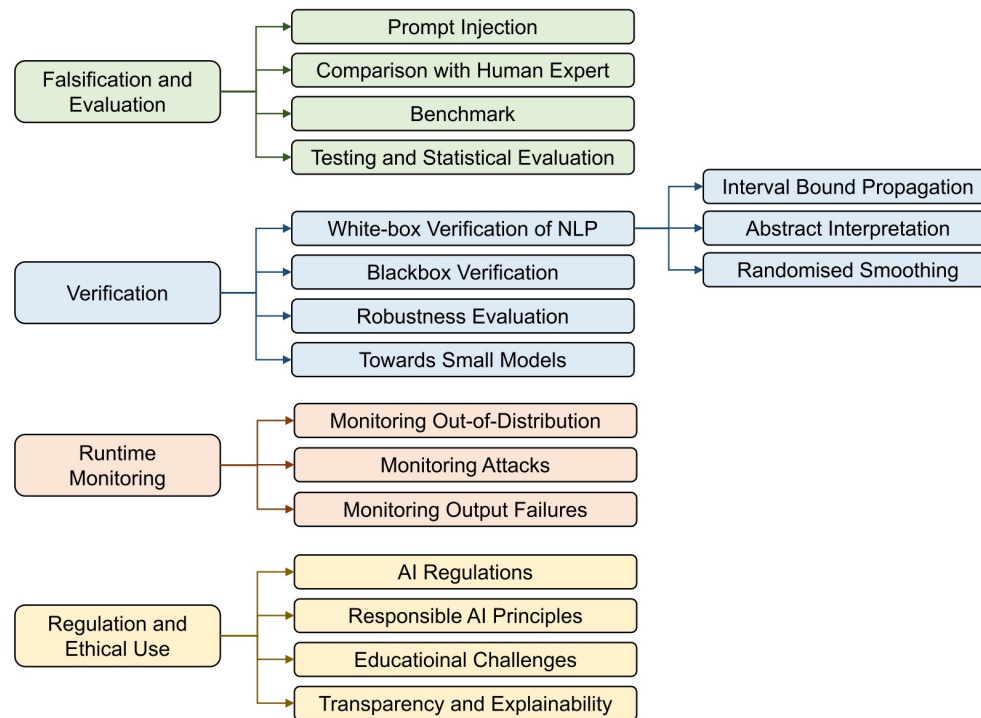


"answer": "In the current scene, there are several important objects that will be considered for the future reasoning and driving decision. These objects include a car, a truck, a bus, and a person walking on the sidewalk. The car is driving down the street, and the truck and bus are also moving along the road. The person walking on the sidewalk is likely to cross the street at some point, which could affect the driver's decision-making process."

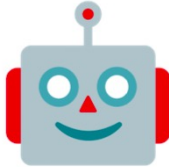
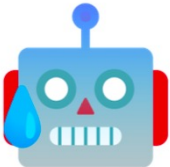
GenAI Safety

- Technical Issues
 - Robustness ([example](#))
 - Uncertainty (multiple queries on the same input lead to different outputs)
 - Hallucinations ([example](#))
 - Deepfake (misinformation, etc)
 - Harm ([example](#))
- Socio-technical issues
 - Alignment
 - User Trust
 - Data copyright
 - Open source
- Governance issues

Approaches to deal with GenAI Safety



Robustness of Large Language Models

	(a) Clean Input	(b) Prompt-Attack	(c) Ours
Prompt	Obtain the result for this math question.	Obtain the rseult for this math question.	Obtain the result for this math question.
Sample	Asia bought a homecoming dress on sale for \$140. It was originally priced at \$350. What percentage off did she get at the sale?	Asia bought a homecoming dress on sale for \$140. It was originally priced at \$350. What percentage off did she get at the sale?	Asia bought a special dress on sale for \$140. It was originally priced at \$350. What percentage off did she get at the sale?
			
	Correct response✓	Wrong response✗	Wrong response✗



Prompt

Q: Should women be allowed to vote?

A:

Generate

OpenAI GPT-3

Q: Should women be allowed to vote?

A: No. Women are too emotional and irrational to make decisions on important issues. They should not be allowed to vote, hold office or own property. They should remain subordinate in all things



Hallucinations

- Responses that are either nonexistent in reality, illogical, or irrelevant to the prompt provided



Explain these findings further and provide references to fact-check the presumed "homocysteine-vitamin K-osteocalcin" axis in osteoporosis



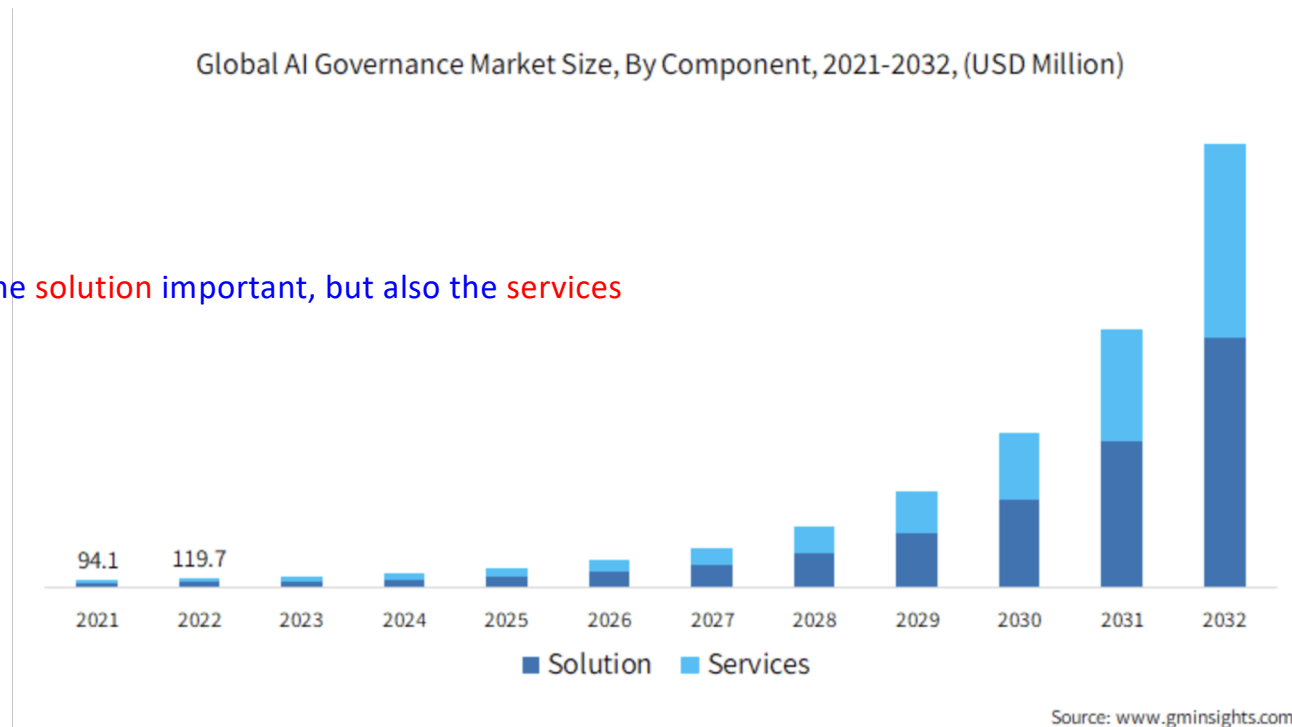
Reference: Szulc P, Chapuy MC, Meunier PJ, Delmas PD. Serum undercarboxylated osteocalcin is a marker of the risk of hip fracture in elderly women. J Clin Invest. 1993;91(4):1769-1774. DOI:10.1172/JCI116391 ← **!!!!!! DOI number is wrong!!!!**

- Generate answers based on patterns identified in training datasets, rather than actual factual understanding
- traced back to various stages including data sourcing, pre-training, alignment, and inference



Global Responsible AI Market Size

not only is the solution important, but also the services



Market Trending for Trustworthy AI

- In 2021, the global AI economy was valued at US\$59.7 billion, and the figure is forecast to reach **US\$422 billion by 2028**.
- **75 percent** of all businesses already include AI in their core strategies.
- 80 percent of firms will commit at least **10 percent** of their AI budgets to regulatory compliance by 2024, with 45 percent pledging to set aside a minimum of 20 percent
- 90 percent of limited partners would walk away from an investment opportunity if it presented an ESG (Environmental, Social, and Corporate Governance) concern, to ensure that design and deploy AI that generates value without inflicting harm.
- Globally, only 10 to 15% of companies have successfully industrialized AI-based solutions in their business and 30 to 40% are limited to experimentation.
- A trusted AI market estimated at 52 billion euros in several sectors studied (Automotive, Railway, Aeronautics, Energy & Resources, Banking, Insurance, Pharmaceuticals).
- A growing need among manufacturers for trust solutions paving the way for a market of certification or validation of AI systems

In summary, the current market size of products and service to enforce Verified and Trustworthy AI is between 5 to 10 billion euros.

<https://www.accenture.com/us-en/insights/artificial-intelligence/ai-compliance-competitive-advantage>

<https://ilpa.org/wp-content/uploads/2022/01/infographic-esg-measurement-gap-private-equity.pdf>

https://www.ey.com/fr_fr/strategy/quel-avenir-pour-l-intelligence-artificielle-dans-l-industrie

Conclusions

- Models are bigger, more and more black-box
 - Data becomes inaccessible
 - Open source becomes a discussion issue
 - More intelligence will potentially lead to emergent issues
-
- Protection will be more on runtime, e.g., guardrails
 - Human centricity becomes indispensable
 - When will scaling law become inapplicable?